umble

Project Report

1. Readme

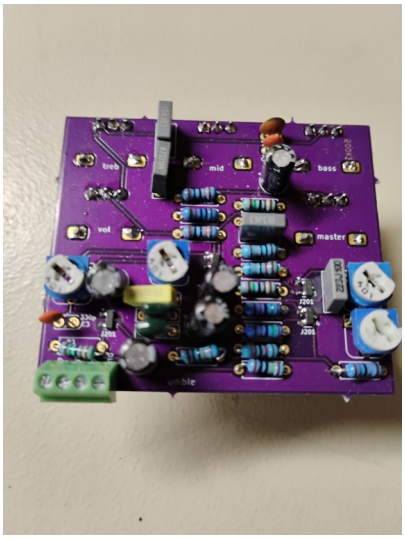
This is from Runoffgrove Umble, and is based on that schematic.

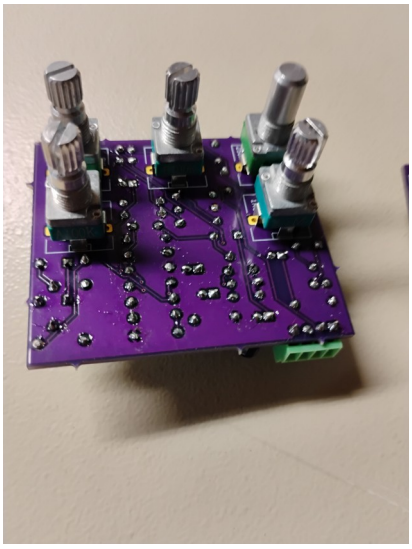
This pedal is verified. It is very loud. You must adjust the biases to around 4.5v on pin3 of each trimmer.

Out of an enclosure it has a lot of noise, which I think is common for higher-gain pedals. All 5 controls seem to work the way that they are intended.

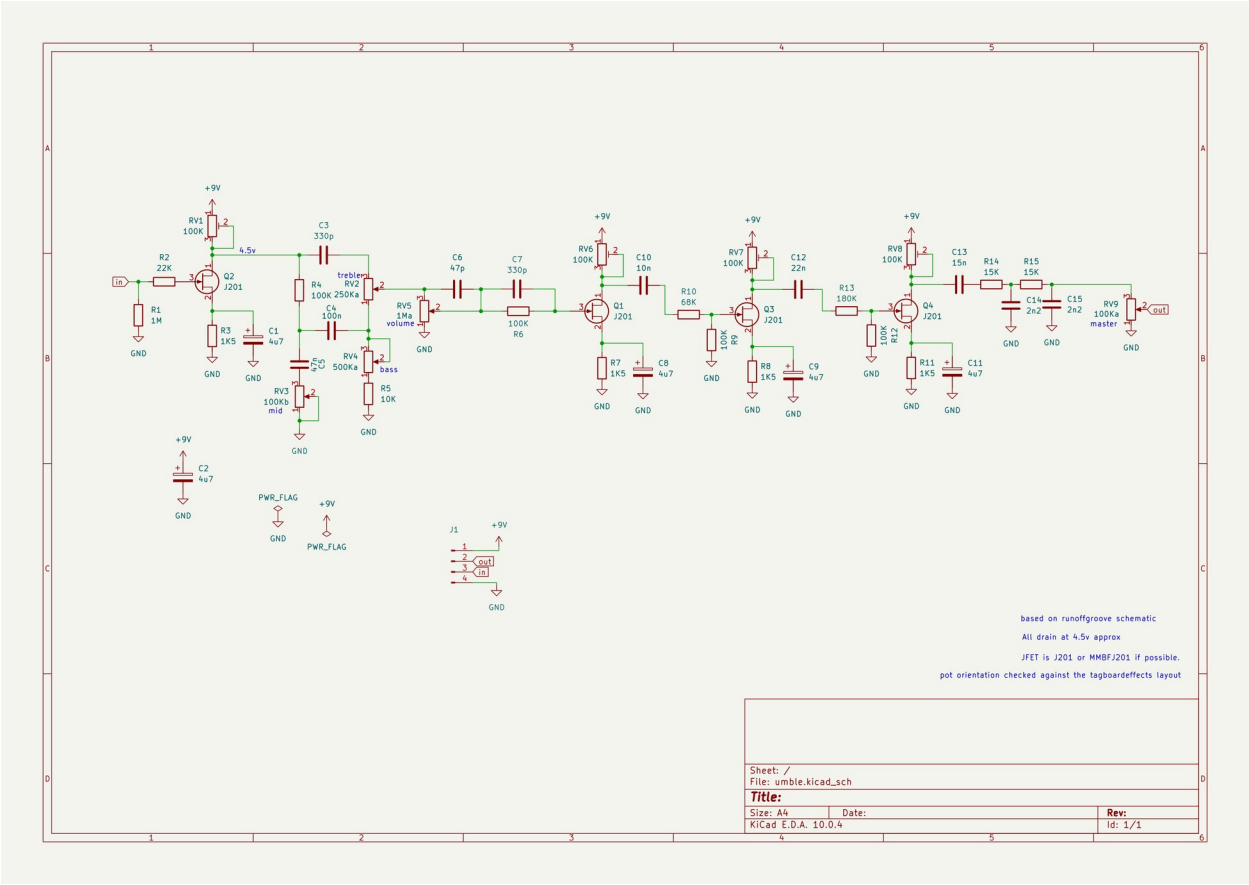
This only has footprints for SMD because I did not think there was enough room for a through-hole and SMD. There probably was enough space to include a bigger footprint to include through-hole. Oh well. I soldered on the J201 first, and then used other pads on the empty pcb to test that the soldered J201s were functional. All 4 tested out as NJFET with a peak tester.

Sourcing all the pots with a RG097 footprint was a bit of a PITA, but I do love how tight these pots are when it is all boxed up. Not boxed yet, tho.

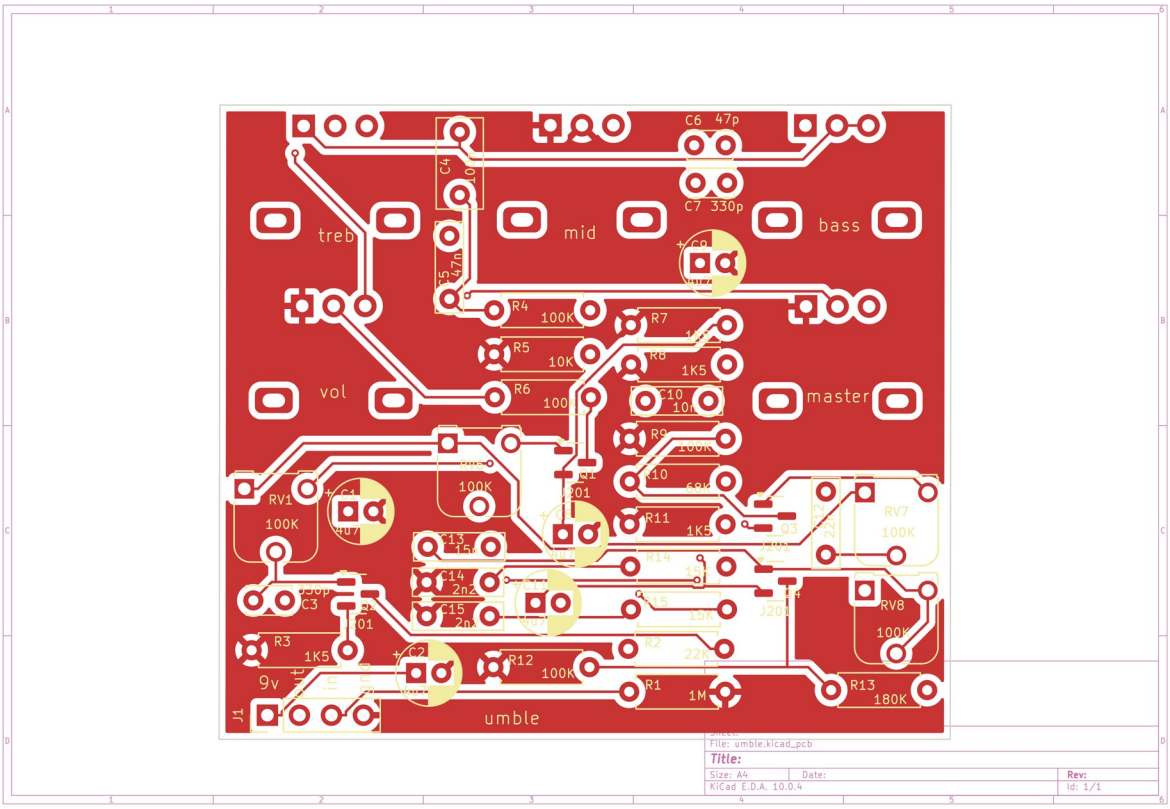




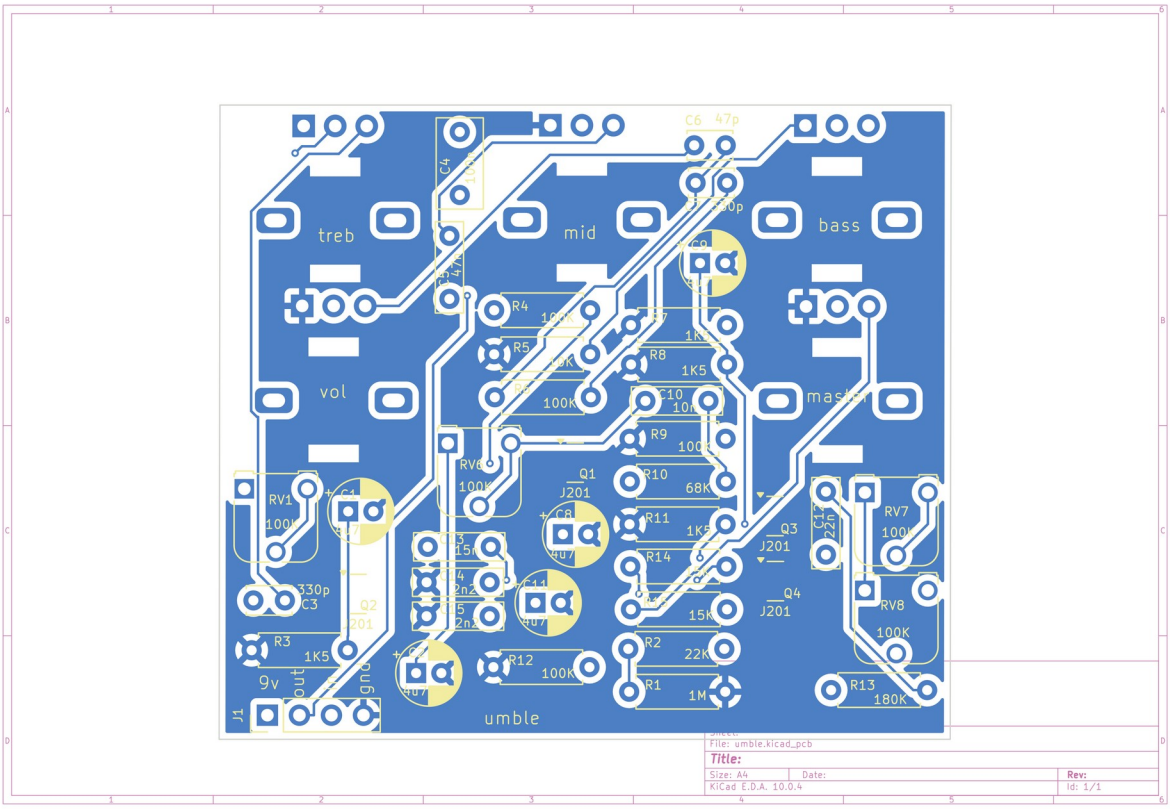
2. Schematic


+9V
RV1 100K
4.5v
R2 22K
Q2 J201
in
R1 1M
R3 1K5
C1 4u7
C3 330p
R4 100K
C4 100n
RV3 100Kb mid
RV2 250Ka treble
RV4 500Ka bass
R5 10K
C6 47p
C7 330p
RV5 1Ma volume
R6 100K
RV6 100K
C10 10n
Q1 J201
R7 1K5
C8 4u7
R10 68K
RV7 100K
C12 22n
Q3 J201
R8 1K5
C9 4u7
R9 100K
R13 180K
RV8 100K
C13 15n
R14 15K
R15 15K
Q4 J201
R12 100K
R11 1K5
C11 4u7
C14 2n2
C15 2n2
RV9 100Ka master
out
C2 4u7
PWR_FLAG
J1
GND
based on runoffgroove schematic
All drain at 4.5v approx
JFET is J201 or MMBFJ201 if possible.
pot orientation checked against the tagboardeffects layout
Sheet: /
File: umble.kicad_sch
Title:
Size: A4
Date:
Rev:
KiCad E.D.A. 10.0.4
Id: 1/1

3. PCB F.Cu + Silkscreen



4. PCB B.CU + Silkscreen



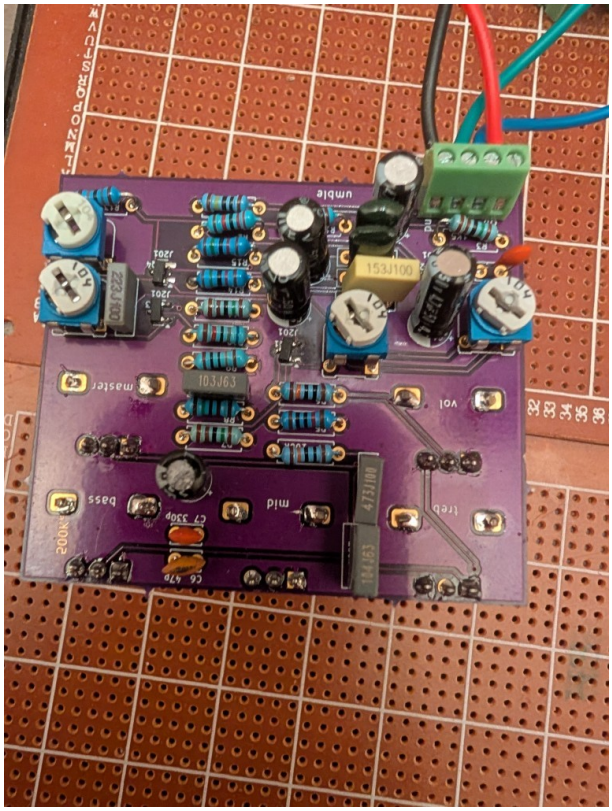
5. Bill of Materials

Reference	Value	Footprint	Qty	DNP
C1,C2,C8,C9,C11	4u7	Capacitor_THT:CP_Radial_D5.0mm_P2.00mm	5	
C3,C7	330p	Capacitor_THT:C_Disc_D3.4mm_W2.1mm_P2.50mm	2	
C4	100n	Capacitor_THT:C_Rect_L7.0mm_W3.5mm_P5.00mm	1	
C5	47n	Capacitor_THT:C_Rect_L7.0mm_W2.0mm_P5.00mm	1	
C6	47p	Capacitor_THT:C_Disc_D3.4mm_W2.1mm_P2.50mm	1	
C10	10n	Capacitor_THT:C_Rect_L7.0mm_W2.0mm_P5.00mm	1	
C12	22n	Capacitor_THT:C_Rect_L7.0mm_W2.0mm_P5.00mm	1	
C13	15n	Capacitor_THT:C_Rect_L7.0mm_W2.0mm_P5.00mm	1	
C14,C15	2n2	Capacitor_THT:C_Rect_L7.0mm_W2.0mm_P5.00mm	2	
J1	Conn_01x04_Pin	Connector_PinHeader_2.54mm:PinHeader_1x04_P2.54mm_Vertical	1	
Q1-Q4	J201	Package_TO_SOT_SMD:SOT-23	4	
R1	1M	Resistor_THT:R_Axial_DIN0207_L6.3mm_D2.5mm_P7.62mm_Horizontal	1	
R2	22K	Resistor_THT:R_Axial_DIN0207_L6.3mm_D2.5mm_P7.62mm_Horizontal	1	
R3,R7,R8,R11	1K5	Resistor_THT:R_Axial_DIN0207_L6.3mm_D2.5mm_P7.62mm_Horizontal	4	
R4,R6,R9,R12	100K	Resistor_THT:R_Axial_DIN0207_L6.3mm_D2.5mm_P7.62mm_Horizontal	4	
R5	10K	Resistor_THT:R_Axial_DIN0207_L6.3mm_D2.5mm_P7.62mm_Horizontal	1	
R10	68K	Resistor_THT:R_Axia	1	

		l_DIN0207_L6.3mm_D2.5mm_P7.62mm_Horizontal		
R13	180K	Resistor_THT:R_Axial_DIN0207_L6.3mm_D2.5mm_P7.62mm_Horizontal	1	
R14,R15	15K	Resistor_THT:R_Axial_DIN0207_L6.3mm_D2.5mm_P7.62mm_Horizontal	2	
RV1,RV6-RV8	100K	Potentiometer_THT:Potentiometer_Runtron_RM-065_Vertical	4	
RV2	250Ka	Potentiometer_THT:Potentiometer_Alps_RK09L_Single_Vertical	1	
RV3	100Kb	Potentiometer_THT:Potentiometer_Alps_RK09L_Single_Vertical	1	
RV4	500Ka	Potentiometer_THT:Potentiometer_Alps_RK09L_Single_Vertical	1	
RV5	1Ma	Potentiometer_THT:Potentiometer_Alps_RK09L_Single_Vertical	1	
RV9	100Ka	Potentiometer_THT:Potentiometer_Alps_RK09L_Single_Vertical	1	

Total line items: 25

pics:


master
vol
bass
mid
treb
103J63
153J100
LM1875T
C6 470
C7 330P
200K

